Practice Problems

Dataset : https://drive.google.com/file/d/1WeHEudireTIGtW7u5HW6w0C3FVMkwC1T/view?usp=sharing

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1. Add New Column and Save File

You have a DataFrame with column `Price`.
Create a new column `Discounted_Price with` (10% discounted price).
Save the modified DataFrame as `discounted_products.csv`.

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2. Check Duplicates and Remove Them

Column: `EmployeeID`
Tasks:

- Count duplicate entries
- Remove duplicates
- Display cleaned DataFrame

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3. Handle Missing Values

Columns: `Age`, `Salary`
Tasks:

- Count missing values
- Fill missing Age with mean age
- Drop rows where `Salary` is missing

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4. Apply Statistical Functions

Columns: `Marks1`, `Marks2`, `Marks3`
Tasks:

- Calculate mean, max for each subject

- Compute total marks per student
- Find standard deviation of total marks

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5. Histogram, Bar, Scatter, Pie

Create:

- Histogram of Age
- Scatter plot (Age vs Salary)
- Scatter plot (Price vs Discounted Price)

6.
Dataset :https://drive.google.com/file/d/1Sy7rA0G9gTmeltVZ-XlugkKF4CPPqvxU/view?usp=sharing

1. Load the dataset from the URL into a pandas DataFrame.
2. Create a scatter plot with `Study_Hours` on the X-axis and `IQ` on the Y-axis.
3. Create a scatter plot with Shoe_Size on the X-Axis and Study_Hours on the Y-axis.
4. Create a histogram of Shoe_Size
5. Add a title and axis labels to make the plot clear.
6. After you plot it, answer the question: Does there appear to be a correlation?